

[11] TurboRAG: Accelerating Retrieval-Augmented Generation with Precomputed KV Caches for Chunked Text

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1-Minute Executive Takeaway

Key Finding: Proves that avoiding repeated text encoding is the key to high-throughput RAG systems.

Core Research Objective

To accelerate RAG response generation by offline precomputing and caching Key-Value states for static knowledge base chunks.

Methodology & Architecture

Pre-processes text chunks through LLM attention layers offline and stores KV tensors, retrieving and concatenating KV representations at query time to bypass token prefill.

Key Empirical Findings & Evidence

TurboRAG reduced prompt processing latency by up to 80% with minimal degradation in generation accuracy.

Direct Relevance to Our Kwiz Paper

Cited in Section 2.4 and Section 6.3 alongside CacheBlend. We cite TurboRAG to contextualize where caching occurs in modern RAG literature and clarify that Kwiz's contribution lies in workload-level courseware lifecycle reuse rather than algorithmic KV-cache manipulation.